AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I111*

Team Name: *LARPSTACK*

Domain: *Advanced Materials, Manufacturing Industrial Innovation*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Problem Statement

Industrial factories rely heavily on CCTV cameras, yet conventional surveillance provides limited spatial awareness of worker and vehicle movements. The lack of unified, real-time spatial information makes it difficult to identify potential safety risks, monitor congestion, and understand operational inefficiencies. FactoryFlow addresses this gap by transforming existing CCTV feeds into a spatially aware Digital Twin that enables real-time object tracking, factory-floor localization, and actionable operational insights.

Problem Understanding

- **Who experiences this problem?** Factory workers, safety officers, production managers, and factory operators.
- **What causes the problem?** Conventional CCTV provides video feeds without unified spatial awareness of worker movements, vehicle locations, and factory-floor activities.
- **How does it affect users or stakeholders?** Delayed hazard identification, limited visibility into congestion and movement patterns, and difficulty making informed safety and operational decisions.

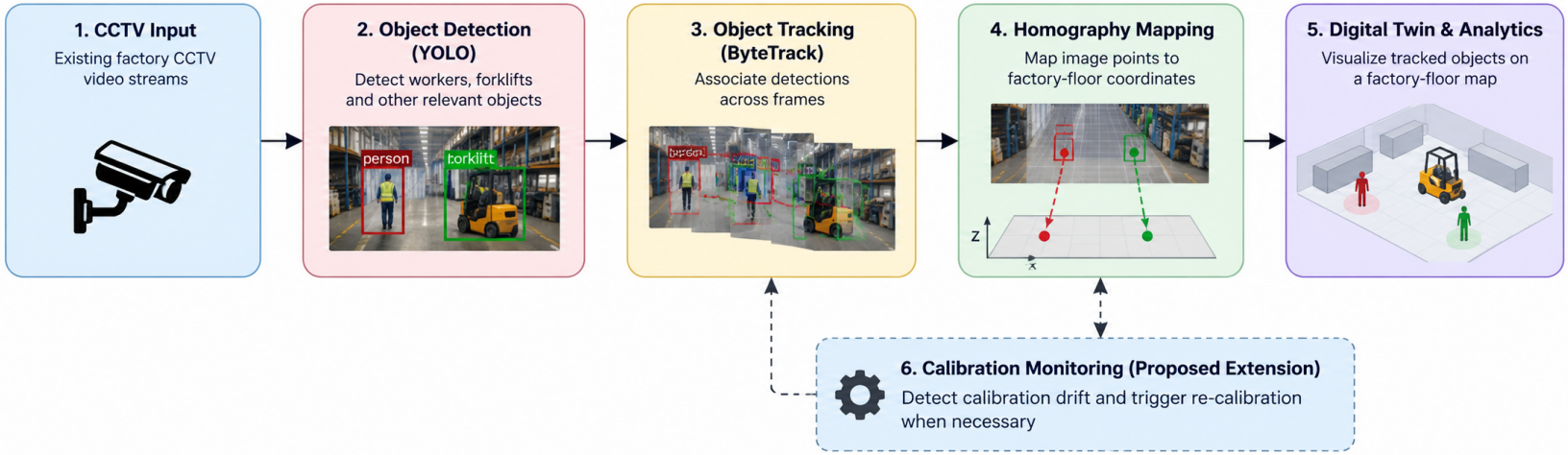
Existing Gaps & Challenges

- **Limited spatial awareness**: Conventional CCTV provides video feeds without real-world factory-floor coordinates.
- **Fragmented operational insights**: Worker movements, vehicle proximity, and congestion are difficult to analyze across the factory floor.
- **Calibration reliability**: Camera movement can invalidate spatial mapping, causing inaccurate location estimates and unreliable downstream analytics.

Our Proposed Solution

- **Core Idea**: FactoryFlow transforms existing factory CCTV into a spatially aware Digital Twin.
- **How It Works**: YOLO detects workers and forklifts, ByteTrack tracks their movements, and homography maps their positions onto factory-floor coordinates.
- **Value Delivered**: Enables spatial monitoring, potential hazard identification, congestion analysis, and improved operational visibility, with adaptive calibration monitoring proposed to address camera drift.

Technical Approach



CCTV footage is processed to detect and track workers and forklifts. Homography maps their estimated ground-contact points to factory-floor coordinates, enabling spatial visualization and operational insights. Calibration-drift monitoring is a proposed extension.

Tech Stack

Frontend: React, TypeScript, Three.js

Backend: Python, FastAPI

Database: Spatial-state storage

AI/ML, APIs, or Hardware: YOLO, ByteTrack, OpenCV, Homography

Innovation & Uniqueness

- **What is new or different?**
  Transforms conventional CCTV into a spatially aware Digital Twin, with proposed adaptive camera-calibration monitoring.

- **Why is this approach suited to the problem?**
  Leverages existing CCTV infrastructure to track movements and map objects onto factory-floor coordinates.

- **What is the key differentiator?**
  Combines object tracking, spatial mapping, and proposed calibration-drift detection in one system.

Feasibility & Scalability

- **Feasibility:** Uses existing CCTV feeds, YOLO, ByteTrack, homography-based mapping, and a React–Three.js Digital Twin with a Python FastAPI backend.
- **Scalability:** Can expand to additional cameras and factory zones through modular processing, unified floor coordinates, and optimized video inference.
- **Risks:** Occlusion, camera movement, calibration errors, and processing load may affect reliability; mitigate through tracking, calibration checks, and inference optimization.

Cost for Product (Optional)

- **Development or hardware cost:** Existing development laptop and CCTV infrastructure; no dedicated hardware required for the initial prototype.
- **Hosting or maintenance cost:** Minimal for local deployment; additional compute, storage, and maintenance costs depend on the number of cameras and deployment scale.
- **Estimated total:** Prototype cost depends on existing infrastructure and available computing resources; a deployment budget is yet to be determined.

Use Cases (compulsorily 3 use cases must be there)

- **Use case 1: Worker–Forklift Safety**
  Safety officers monitor worker and forklift proximity to identify potentially hazardous situations.

- **Use case 2: Congestion and Workflow Analysis**
  Production managers analyze movement patterns, zone occupancy, and bottlenecks to improve factory-floor efficiency.

- **Use case 3: Camera Calibration Monitoring**
  Factory operators detect potential calibration drift and initiate recalibration to maintain reliable spatial mapping.

Impacts and Benefits

- **Benefit to target users:** Enables safety officers and production managers to monitor worker–forklift proximity, visualize movement, and identify potential congestion.

- **Social, economic, or environmental impact:** Supports workplace safety, improves operational visibility, and leverages existing CCTV infrastructure to reduce additional sensing needs.

- **Expected measurable outcome:** Evaluate localization error, object-tracking performance, congestion indicators, and calibration-drift detection using controlled tests.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 8	Target 8.8	Decent Work and Economic Growth	Supports safer working environments through worker–vehicle proximity monitoring.
SDG 9	Target 9.4	Industry, Innovation and Infrastructure	Uses existing CCTV infrastructure to enable data-driven industrial monitoring.

Individual Member Contribution

Member	Contribution or responsibility
Akchara P	Defined the problem scope; researched computer vision approaches and YOLO-based object detection for industrial environments.
Adhithyan A S	Researched multi-object tracking techniques, movement analysis, and worker–forklift safety monitoring.
Kavin Prasath A R	Studied homography-based spatial mapping, camera calibration, and factory-floor localization methods.
Divya Sri K	Researched Digital Twin architectures, system design, visualization technologies, and backend integration.

Yukti ID

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